Lesson 3.16 - How to Calculate the Median of Grouped Data

> **Video:** How To Calculate the Median of Grouped Data by The Organic Chemistry Tutor **Chapter:** 3 - Central Tendency

Formula

$$\text{median} = L + \left(\frac{\frac{n}{2} - C}{f}\right) h$$

Symbol	Meaning
L	**Lower boundary** of the median class
n	Total frequency, $\sum f$
C	**Cumulative frequency of the class before** the median class
f	Frequency of the median class
h	Class width, **upper boundary − lower boundary**

Class boundaries vs class limits

If the classes are written 60-69, 70-79, ..., there is a gap between 69 and 70. The **boundaries** sit halfway across each gap:

- lower boundary of 70-79 = $\dfrac{69 + 70}{2} = 69.5$
- upper boundary of 70-79 = $\dfrac{79 + 80}{2} = 79.5$
- width $h = 79.5 - 69.5 = 10$, **not** $79 - 70 = 9$

Example

Grade	f	Cumulative f
40-49	3	3
50-59	5	8
60-69	6	**14** ← C
70-79	**9**	23 ← median class
80-89	8	31
90-100	7	38

Step 1 - find the median class. $n = 38$ and $n/2 = 19$. The classes hold the scores in positions 1-3, 4-8, 9-14 and **15-23**, ... The 19th value is in the **70-79** class.

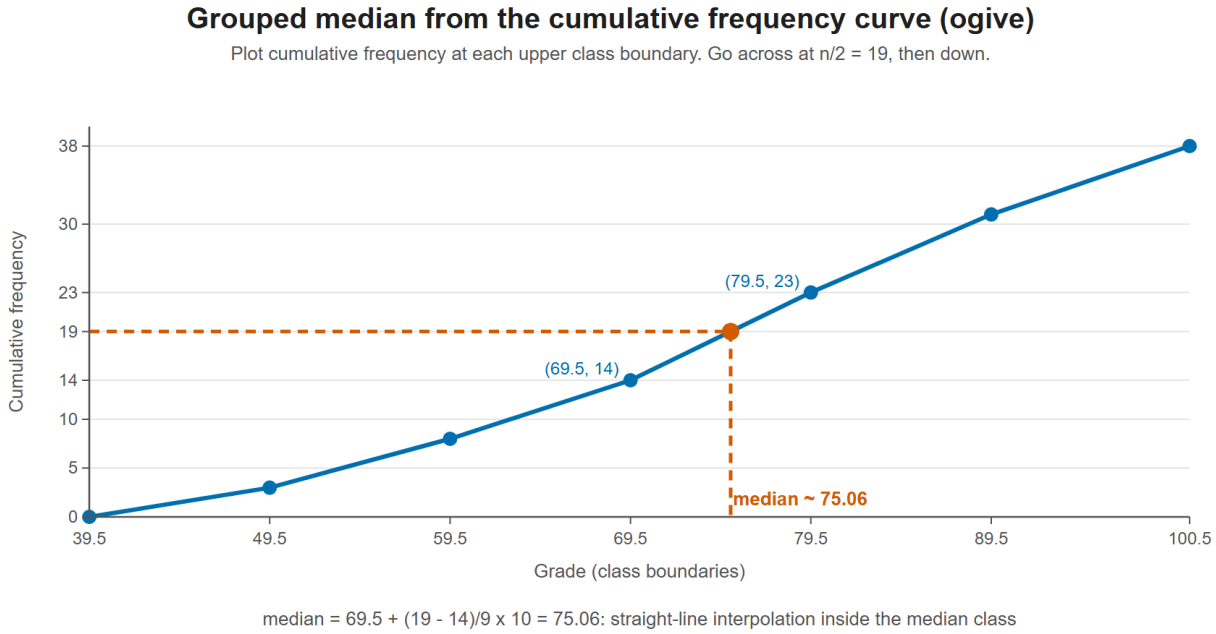
Step 2 - collect the values. $L = 69.5$, $C = 14$, $f = 9$, $h = 10$.

Step 3 - substitute.

$$\text{median} = L + \left(\frac{\frac{n}{2} - C}{f}\right) h = 69.5 + \left(\frac{19 - 14}{9}\right)(10) = 69.5 + \frac{50}{9} \approx 69.5 + 5.56 = 75.06$$

The median is about **75.1**, which is inside the median class 70-79 as it should be.

> **Correction to the video:** the video gives 75.5 (repeating) as the result. But $50/9 \approx 5.56$ and $69.5 + 5.56 \approx 75.06$, so the correct median is about **75.06**.



The formula is the same as reading the cumulative frequency curve. Go across from n/2 = 19 to the curve, then straight down. The median lands about 5/9 of the way through the 69.5-79.5 class.

Key takeaways

- Find the **median class** first, using the cumulative frequency to locate position $n/2$.
- Use class **boundaries** (69.5, 79.5) for L and h when there are gaps between class limits.
- C is the cumulative frequency **before** the median class, not including it.
- Check that the answer lies inside the median class.